

Education

The University of Texas at Austin

Master of Science, Computer Science

Austin, TX

August 2026 – Expected May 2028

University of Michigan

Bachelor of Science in Engineering, Computer Science | GPA: 3.67

Ann Arbor, MI

August 2022 – December 2025

- Magna Cum Laude

Experience

KBR, Inc.

Ann Arbor, MI

Software Engineer | C++, MATLAB, Python

January 2026 – Present

- Develop and integrate mission-critical radar software for defense and intelligence systems across signal processing, machine learning, simulation, and computer vision
- Optimize radar application performance by streamlining data-processing, software-engineering, and algorithmic workflows in an agile development environment

Software Engineer Intern | MATLAB, Python

May 2024 – August 2024; May 2025 – August 2025

- Refactored and migrated a Synthetic Aperture Radar (SAR) Automatic Target Recognition (ATR) framework from TensorFlow to MATLAB using the Deep Learning Toolbox, improving maintainability and streamlining deployment
- Developed and validated radar and geospatial utilities for LLA, ECEF, ENU, and UTM coordinate transformations and 3D visualization of beam patterns, iso-range spheres, iso-Doppler cones, velocity vectors, and angular geometry
- Built image-processing tools to suppress spurious artifacts and reconnect discontinuous regions in binary images

Harman International

Novi, MI

System Test Engineer Intern | Arduino, C#, HTML/CSS, PHP, SQL

May 2023 – August 2023

- Spearheaded the Harman Phone Management System, an internal platform for storing, securing, and managing engineering test devices, integrating an Arduino-controlled deadbolt with a web interface and administrative tooling
- Strengthened platform security by implementing HTTPS with OpenSSL-generated SSL certificates and PHP session-based authentication for secure user and database management

Projects

Secure AI Gateway | Python, FastAPI, PostgreSQL, Redis, Kubernetes

- Built an OpenAI-compatible LLM security gateway with API-key authentication, model authorization, distributed rate limiting, usage budgets, PII and prompt-injection controls, and least-privilege tool authorization
- Engineered PostgreSQL/Redis-backed authoritative state, hardened Docker/Kubernetes deployment, Prometheus/OpenTelemetry observability, and adversarial, resilience, security, and supply-chain verification in CI

Detecting Dirty Dish Culprits | Python, YOLOv8, Roboflow, OpenCV

- Built a YOLOv8-based computer vision pipeline to classify washing versus walking behavior in kitchen footage and flag likely dirty-dish drop-offs
- Annotated custom video data, added temporal continuity logic, and evaluated the detector with a 70/20/10 split, achieving 0.96 mAP50

Decrypting Blockchain: The Art of the Steal | Cryptography, Blockchain, Security Analysis

- Researched Bitcoin and Monero security/privacy tradeoffs, analyzing ring signatures, RingCT, stealth addresses, Dandelion++, traceability limits, and scam case studies

Bisdorf.org | Astro, TypeScript, CSS, Cloudflare Pages

- Built and deployed a responsive portfolio site with reusable Astro layouts, static project data, resume rendering/download support, project pages, and search indexing

SillyQL | C++, Hash Tables, Binary Search Trees, Query Processing

- Built a simplified SQL-style database engine using hash tables and binary search trees to support command-driven query processing and table joins

Network Security | Python, Network Security, Protocol Analysis, Packet Handling

- Implemented security-focused networking analyses with Python packet-handling tooling, evaluating protocol behavior, attacks, and defenses